

# JOONWON LEE

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## SUMMARY

Statistics Ph.D. candidate and quantitative researcher with experience in stochastic modeling, likelihood-based inference, statistical diagnostics, model validation, and large-scale data processing. Builds reproducible Python/PyTorch workflows for noisy, irregular dependent data, integrates compiled C++ components where needed, and investigates the root causes of unexpected model and data behavior.

## EDUCATION

**Ph.D. in Statistics, Rutgers University, Piscataway, NJ**

Expected Jan 2027 | GPA: 3.8/4.0

**M.S. in Statistics, University of Minnesota, Minneapolis, MN**

Sep 2021 | GPA: 3.7/4.0

## TECHNICAL SKILLS

**Programming and Computing:** Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), C++/pybind11 integration, SQL, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing and restartable data/modeling pipelines

**Quantitative Methods:** Stochastic processes, likelihood-based inference, Gaussian processes, Monte Carlo simulation, numerical optimization, regression and machine learning, hypothesis testing, spectral analysis, model diagnostics and validation

## EXPERIENCE

**JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review**

Summer 2026

- Developed and validated a Sequential Probability Ratio Test framework for binary risk indicators, translating a monitoring problem into explicit hypotheses, likelihood-ratio decision boundaries, and statistically controlled stopping rules.
- Applied Brownian-motion approximation and dynamic-programming probability propagation to quantify boundary-crossing probabilities, stopping-time behavior, Type I/II error, and early-decision tradeoffs.

**Travelers - Data Science Leadership Program, Business Insurance Pricing Team**

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model validation, and performance diagnostics on AWS EC2.
- Presented model diagnostics and feature-attribution findings as pricing and risk-segmentation recommendations for business stakeholders.

## QUANTITATIVE RESEARCH AND TRADING

**Ph.D. Dissertation Research - Scalable Stochastic Modeling and Diagnostics**

Sep 2024 - Present

- **Scalable Inference:** Developed an advection-aware Vecchia approximation that factorizes a large Gaussian-process likelihood into ordered local conditional models, replacing dense covariance operations to fit up to 145,008 observations per day.
- **Data Processing and Root-Cause Analysis:** Identified systematic measurement geometry, including a cross-scan latitude slope with distance from instrument nadir and hour-specific westward grid drift; applied time-specific longitude offsets, quality filtering, missingness tracking, and KD-tree center matching.
- **Inference-Driven Diagnostics:** Designed scale- and frequency-resolved tools to pinpoint whether model misspecification arises from low-frequency structure, high-frequency behavior, or space-time dependence.
- **Research Engineering:** Integrated compiled C++ max-min ordering routines into Python through pybind11 and built reusable Python/PyTorch workflows with CPU/GPU execution, cached artifacts, and restartable HPC jobs.

**IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants**

Apr 2025

- Built and evaluated a market-making strategy combining market-based fair-value estimation with inventory-aware quoting and rolling-window soft/hard liquidation controls.

## ADDITIONAL

**Coursework:** Probability Theory, Stochastic Processes, Machine Learning, Linear Algebra, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Econometrics

**Languages:** Korean (Native); English (Fluent); Chinese (Basic)